

Object Oriented Programming Using C++

LECTURE NOTES ON Object Oriented Programming Using C++

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SYLLABUS

PCCS2207 Object Oriented Programming

Module I

Introduction to object oriented programming, user defined types, structures, unions, polymorphism, encapsulation. Getting started with C++ syntax, data-type, variables, strings, functions, default values in functions, recursion, namespaces, operators, flow control, arrays and pointers.

Module II

Abstraction mechanism: Classes, private, public, constructors, destructors, member data, member functions, inline function, friend functions, static members, and references.

Inheritance: Class hierarchy, derived classes, single inheritance, multiple, multilevel, hybrid inheritance, role of virtual base class, constructor and destructor execution, base initialization using derived class constructors.

Polymorphism: Binding, Static binding, Dynamic binding, Static polymorphism: Function Overloading, Ambiguity in function overloading, Dynamic polymorphism: Base class pointer, object slicing, late binding, method overriding with virtual functions, pure virtual functions, abstract classes.

Operator Overloading: This pointer, applications of this pointer, Operator function, member and non member operator function, operator overloading, I/O operators.

Exception handling: Try, throw, and catch, exceptions and derived classes, function exception declaration, unexpected exceptions, exception when handling exceptions, resource capture and release.

Module III

Dynamic memory management, new and delete operators, object copying, copy constructor, assignment operator, virtual destructor.

Template: template classes, template functions.

Standard Template Library: Fundamental idea about string, iterators, hashes, iostreams and other types.

Namespaces: user defined namespaces, namespaces provided by library. Object Oriented Design, design and programming, role of classes.

Text Books:

1. Object Oriented Programming with C++ by E. Balagurusamy, McGraw-Hill Education (India)
2. ANSI and Turbo C++ by Ashoke N. Kamthane, Pearson Education

Reference Books:

1. Big C++ - Wiley India
2. C++: The Complete Reference- Schildt, McGraw-Hill Education (India)
3. C++ and Object Oriented Programming – Jana, PHI Learning.
4. Object Oriented Programming with C++ - Rajiv Sahay, Oxford
5. Mastering C++ - Venugopal, McGraw-Hill Education (India)

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Module-1:

LECTURE-1

Introduction:

Programmers write instructions in various programming languages to perform their computation tasks such as:

- (i) Machine level Language
- (ii) Assembly level Language
- (iii) High level Language

Machine level Language :

Machine code or machine language is a set of instructions executed directly by a computer's central processing unit (CPU). Each instruction performs a very specific task, such as a load, a jump, or an ALU operation on a unit of data in a CPU register or memory. Every program directly executed by a CPU is made up of a series of such instructions.

Assembly level Language :

An assembly language (or assembler language) is a low-level programming language for a computer, or other programmable device, in which there is a very strong (generally one-to-one) correspondence between the language and the architecture's machine code instructions. Assembly language is converted into executable machine code by a utility program referred to as an assembler; the conversion process is referred to as assembly, or assembling the code.

High level Language :

High-level language is any programming language that enables development of a program in much simpler programming context and is generally independent of the computer's hardware architecture. High-level language has a higher level of abstraction from the computer, and focuses more on the programming logic rather than the underlying hardware components such as memory addressing and register utilization.

The first high-level programming languages were designed in the 1950s. Now there are dozens of different languages, including Ada , Algol, BASIC, COBOL, C, C++, JAVA, FORTRAN, LISP, Pascal, and Prolog. Such languages are considered high-level because they are closer to human languages and farther from machine languages. In contrast, assembly languages are considered low

level because they are very close to machine languages.

The high-level programming languages are broadly categorized in to two categories:

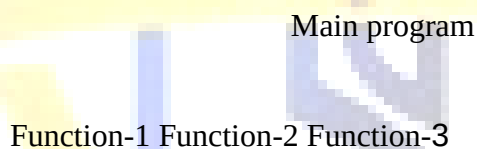
- (iv) Procedure oriented programming (POP) language.
- (v) Object oriented programming (OOP) language.

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Procedure Oriented Programming Language

In the procedure oriented approach, the problem is viewed as sequence of things to be done such as reading, calculation and printing.

Procedure oriented programming basically consist of writing a list of instruction or actions for the computer to follow and organizing these instruction into groups known as functions.



The disadvantage of the procedure oriented programming languages is:

1. Global data access
2. It does not model real word problem very well
3. No data hiding

Global data Global data

Function-1 Function-2 Function-3

Local data Local data Local data

Characteristics of procedure oriented programming:

1. Emphasis is on doing things (algorithm)
2. Large programs are divided into smaller programs known as functions.
3. Most of the functions share global data
4. Data move openly around the system from function to function

5. Function transforms data from one form to another.
6. Employs top-down approach in program design

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LECTURE-2

Object Oriented Programming

“Object oriented programming as an approach that provides a way of modularizing programs by creating partitioned memory area for both data and functions that can be used as templates for creating copies of such modules on demand”.

Object A Object B

Data Data

Communication

Functions Functions

Object C

Functions

Data

Features of the Object Oriented programming

1. Emphasis is on doing rather than procedure.
2. programs are divided into what are known as objects.
3. Data structures are designed such that they characterize the objects.
4. Functions that operate on the data of an object are tied together in the data structure.
5. Data is hidden and can't be accessed by external functions.
6. Objects may communicate with each other through functions.
7. New data and functions can be easily added.
8. Follows bottom-up approach in program design.

LECTURE-3

BASIC CONCEPTS OF OBJECTS ORIENTED PROGRAMMING

1. Objects
2. Classes
3. Data abstraction and encapsulation
4. Inheritance
5. Polymorphism
6. Dynamic binding
7. Message passing

OBJECTS

Objects are the basic run-time entities in an object-oriented system. They may represent a person, a place, a bank account, a table of data or any item that the program must handle.

The fundamental idea behind object oriented approach is to combine both data and function into a single unit and these units are called objects.

The term objects means a combination of data and program that represent some real word entity. For example: consider an example named Amit; Amit is 25 years old and his salary is 2500. The Amit may be represented in a computer program as an object. The data part of the object would be (name: Amit, age: 25, salary: 2500)

The program part of the object may be collection of programs (retrive of data, change age, change of salary). In general even any user –defined type-such as employee may be used. In the Amit object the name, age and salary are called attributes of the object.

Object: Student STUDENT

DATA Total
Name
Date-of-birth
Marks Average

FUNCTIONS
Total
Average Display
Display

CLASS:

A group of objects that share common properties for data part and some program part are collectively called as class.

In C ++ a class is a new data type that contains member variables and member functions that operate on the variables.

DATA ABSTRACTION :

Abstraction refers to the act of representing essential features without including the background details or explanations. Classes use the concept of abstraction and are defined as size, width and cost and functions to operate on the attributes.

DATA ENCAPSALATION :

The wrapping up of data and function into a single unit (called class) is known as encapsulation. The data is not accessible to the outside world and only those functions which are wrapped in the class can access it. These functions provide the interface between the objects data and the program.

INHERITENCE :

Inheritance is the process by which objects of one class acquire the properties of another class. In the concept of inheritance provides the idea of reusability. This means that we can add additional features to an existing class without modifying it. This is possible by designing a new class which will have the combined features of both the classes.

POLYMORPHISM:

Polymorphism means the ability to take more than one form. An operation may exhibit different instances. The behaviour depends upon the type of data used in the operation.

A language feature that allows a function or operator to be given more than one definition. The types of the arguments with which the function or operator is called determines which definition will be used.

Overloading may be operator overloading or function overloading.

It is able to express the operation of addition by a single operator say '+'. When this is possible you use the expression $x + y$ to denote the sum of x and y , for many different types of x and y ; integers, float and complex no. You can even define the $+$ operation for two strings to mean the concatenation of the strings.

DYNAMIC BINDING :

Binding refers to the linking of a procedure call to the code to be executed in response to the call. Dynamic binding means the code associated with a given procedure call is not known until the time of the call at run-time. It is associated with a polymorphic reference depends upon the dynamic type of that reference.

MESSAGE PASSING :

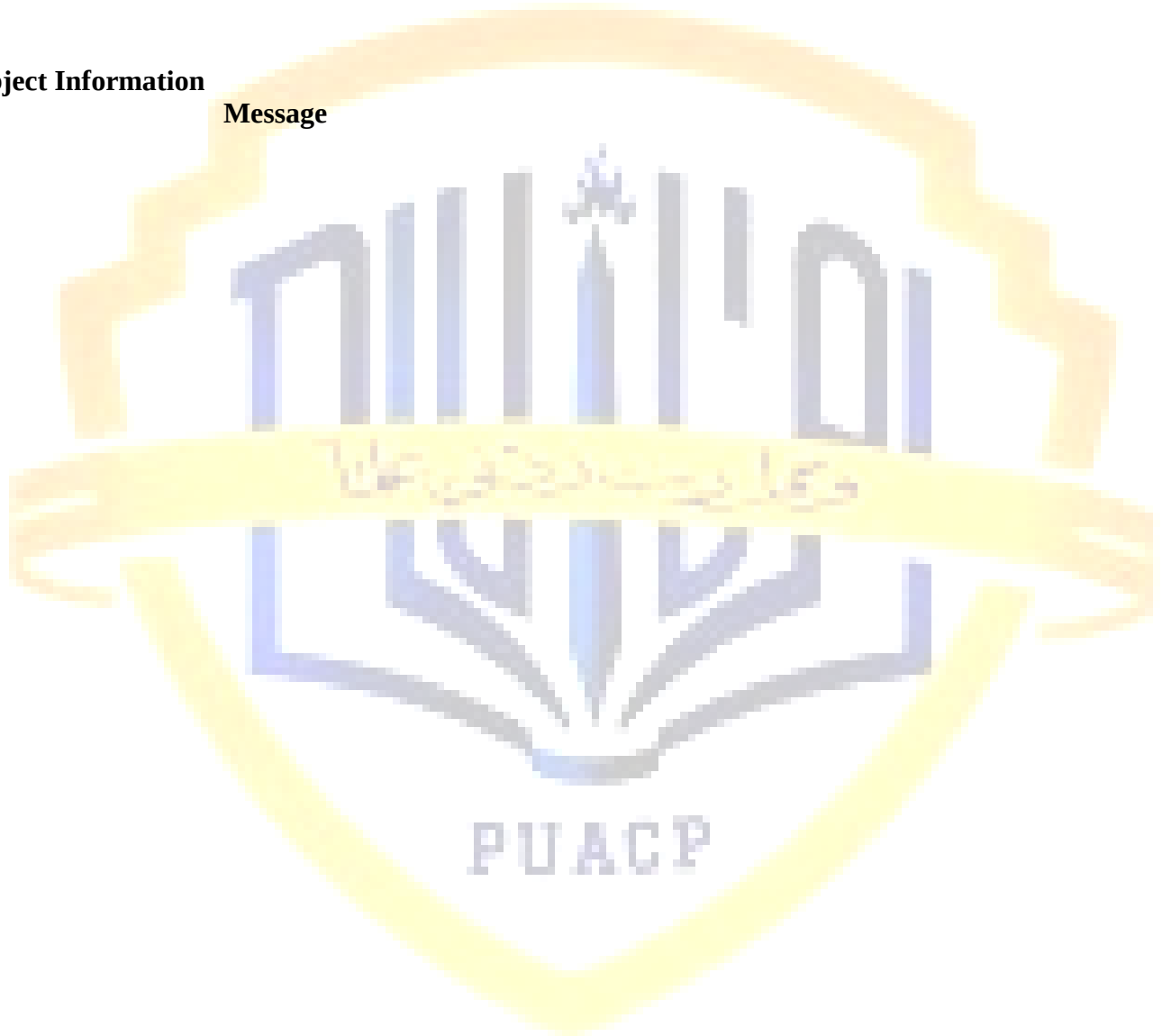
other. An object oriented program consists of a set of objects that communicate with each

A message for an object is a request for execution of a procedure and therefore will invoke a function (procedure) in the receiving object that generates the desired result. Message passing involves specifying the name of the object, the name of the function (message) and information to be sent.

Employee . Salary (name)

Object Information

Message



BENEFITS OF OOP:

Oop offers several benefits to both the program designer and the user. Object-oriented contributes to the solution of many problems associated with the development and quality of software products. The principal advantages are :

1. Through inheritance we can eliminate redundant code and extend the use of existing classes.
2. We can build programs from the standard working modules that communicate with one another, rather than having to start writing the code from scratch. This leads to saving of development time and higher productivity.
3. This principle of data hiding helps the programmer to build secure programs that can't be invaded by code in other parts of the program.
4. It is possible to have multiple instances of an object to co-exist with out any interference.
5. It is easy to partition the work in a project based on objects.
6. Object-oriented systems can be easily upgraded from small to large systems.
7. Message passing techniques for communication between objects makes the interface description with external systems much simpler.
8. Software complexity can be easily managed.

APPLICATION OF OOP:

The most popular application of oops up to now, has been in the area of user interface design such as windows. There are hundreds of windowing systems developed using oop techniques.

Real business systems are often much more complex and contain many more objects with complicated attributes and methods. Oop is useful in this type of applications because it can simplify a complex problem. The promising areas for application of oop includes.

1. Real – Time systems.
2. Simulation and modeling
3. Object oriented databases.
4. Hypertext,hypermedia and expertext.
5. AI and expert systems.
6. Neural networks and parallel programming.
7. Dicision support and office automation systems.
8. CIM / CAM / CAD system.

C ++ is an object oriented programming language, C ++ was developed by Jarney Stroustrup at AT & T Bell lab, USA in early eighties. C ++ was developed from c and simula 67 language. C ++ was early called 'C with classes'.

C++ Comments:

C++ introduces a new comment symbol //(double slash). Comments start with a double slash symbol and terminate at the end of line. A comment may start any where in the line and what ever follows till the end of line is ignored. Note that there is no closing symbol.

The double slash comment is basically a single line comment. Multi line comments can be written as follows:

```
// this is an example of  
// c++ program  
// thank you
```

The c comment symbols /**/ are still valid and more suitable for multi line comments. /* this is an example of c++ program */

Output Operator:

The statement `cout << "Hello, world"` displayed the string with in quotes on the screen. The identifier `cout` can be used to display individual characters, strings and even numbers. It is a predefined object that corresponds to the standard output stream. Stream just refers to a flow of data and the standard Output stream normally flows to the screen display. The `cout` object, whose properties are defined in `iostream.h` represents that stream. The insertion operator `<<` also called the 'put to' operator directs the information on its right to the object on its left.

Return Statement:

In C++ `main ( )` returns an integer type value to the operating system. Therefore every `main ( )` in C++ should end with a `return (0)` statement, otherwise a warning or an error might occur.

Input Operator:

The statement
`cin >> number 1;`

is an input statement and causes. The program to wait for the user to type in a number. The number keyed in is placed in the variable `number1`. The identifier **cin** is a predefined object in C++ that corresponds to the standard input stream. Here this stream represents the key board.

The operator `>>` is known as get from operator. It extracts value from the keyboard and assigns it to the variable on its right.

Cascading Of I/O Operator:

```
cout<<"sum="<<sum<<"\n";  
cout<<"sum="<<sum<<"\n"<<"average="<<average<<"\n";  
cin>>number1>>number2;
```

Structure Of A Program :

Probably the best way to start learning a programming language is by writing a program. Therefore, here is our first program:

```
// my first program in C++
```

```
#include <iostream>
using namespace std;
```

```
int main ()
{
    cout << "Hello World!";
    return 0;
}
```

Output:-Hello World!

The first panel shows the source code for our first program. The second one shows the result of the program once compiled and executed. The way to edit and compile a program depends on the compiler you are using. Depending on whether it has a Development Interface or not and on its version. Consult the compilers section and the manual or help included with your compiler if you have doubts on how to compile a C++ console program.

The previous program is the typical program that programmer apprentices write for the first time, and its result is the printing on screen of the "Hello World!" sentence. It is one of the simplest programs that can be written in C++, but it already contains the fundamental components that every C++ program has. We are going to look line by line at the code we have just written:

// my first program in C++

This is a comment line. All lines beginning with two slash signs (//) are considered comments and do not have any effect on the behavior of the program. The programmer can use them to include short explanations or observations within the source code itself. In this case, the line is a brief description of what our program is.

#include <iostream>

Lines beginning with a hash sign (#) are directives for the preprocessor. They are not regular code lines with expressions but indications for the compiler's preprocessor. In this case the directive `#include<iostream>` tells the preprocessor to include the `iostream` standard file. This specific file (`iostream`) includes the declarations of the basic standard input-output library in C++, and it is included because its functionality is going to be used later in the program.

using namespace std;

All the elements of the standard C++ library are declared within what is called a namespace, the namespace with the name `std`. So in order to access its functionality we declare with this expression that we will be using these entities. This line is very frequent in C++ programs that use the standard library, and in fact it will be included in most of the source codes included in these tutorials. **int**

main ()

This line corresponds to the beginning of the definition of the main function. The main function is the point by where all C++ programs start their execution, independently of its location within the source code. It does not matter whether there are other functions with other names defined before or after it – the instructions contained within this function's definition will always be the first ones to be

executed in any C++ program. For that same reason, it is essential that all C++ programs have a main function.

The word `main` is followed in the code by a pair of parentheses (`()`). That is because it is a function declaration: In C++, what differentiates a function declaration from other types of expressions are these parentheses that follow its name. Optionally, these parentheses may enclose a list of parameters within them.

Right after these parentheses we can find the body of the main function enclosed in braces (`{}`).

What is contained within these braces is what the function does when it is executed. **cout << "Hello World!";**

This line is a C++ statement. A statement is a simple or compound expression that can actually produce some effect. In fact, this statement performs the only action that generates a visible effect in our first program.

cout represents the standard output stream in C++, and the meaning of the entire statement is to insert a sequence of characters (in this case the Hello World sequence of characters) into the standard output stream (which usually is the screen).

cout is declared in the iostream standard file within the std namespace, so that's why we needed to include that specific file and to declare that we were going to use this specific namespace earlier in our code.

Notice that the statement ends with a semicolon character (;). This character is used to mark the end of the statement and in fact it must be included at the end of all expression statements in all C++ programs (one of the most common syntax errors is indeed to forget to include some semicolon after a statement).

return 0;

The return statement causes the main function to finish. return may be followed by a return code (in our example is followed by the return code 0). A return code of 0 for the main function is generally interpreted as the program worked as expected without any errors during its execution. This is the most usual way to end a C++ console program.

You may have noticed that not all the lines of this program perform actions when the code is executed. There were lines containing only comments (those beginning by //). There were lines with directives for the compiler's preprocessor (those beginning by #). Then there were lines that began the declaration of a function (in this case, the main function) and, finally lines with statements (like the insertion into cout), which were all included within the block delimited by the braces ({}) of the main function.

The program has been structured in different lines in order to be more readable, but in C++, we do not have strict rules on how to separate instructions in different lines. For example, instead of

```
int main ()
{
    cout << " Hello World!";
    return 0;
}
```

We could have written:

```
int main ()
{
    cout << "Hello World!";
    return 0;
}
```

All in just one line and this would have had exactly the same meaning as the previous code. In C++, the separation between statements is specified with an ending semicolon (;) at the end of each one, so the separation in different code lines does not matter at all for this purpose. We can write many statements per line or write a single statement that takes many code lines. The division of

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code in different lines serves only to make it more legible and schematic for the humans that may read it.

Let us add an additional instruction to our first program:

```
// my second program in C++
#include <iostream>
using namespace std;
```

```
int main ()
```

```

{
cout << "Hello World! ";
cout << "I'm a C++ program";
return 0;
}

```

Output:-Hello World! I'm a C++ program

In this case, we performed two insertions into cout in two different statements. Once again, the separation in different lines of code has been done just to give greater readability to the program, since main could have been perfectly valid defined this way:

```

int main ()
{
cout << " Hello World! ";
cout << " I'm a C++ program ";
return 0;
}

```

We were also free to divide the code into more lines if we considered it more convenient:

```

int main ()
{
cout << "Hello World!";
cout << "I'm a C++ program";
return 0;
}

```

And the result would again have been exactly the same as in the previous examples. Preprocessor directives (those that begin by #) are out of this general rule since they are not statements. They are lines read and processed by the preprocessor and do not produce any code by themselves. Preprocessor directives must be specified in their own line and do not have to end with a semicolon (;).

STRUCTURE OF C++ PROGRAM

- **Include files**
- **Class declaration**
- **Class functions, definition**
- **Main function program**

Example :-

```

# include<iostream.h>

class person

{
char name[30];
int age;
public:
void getdata(void);

```

```

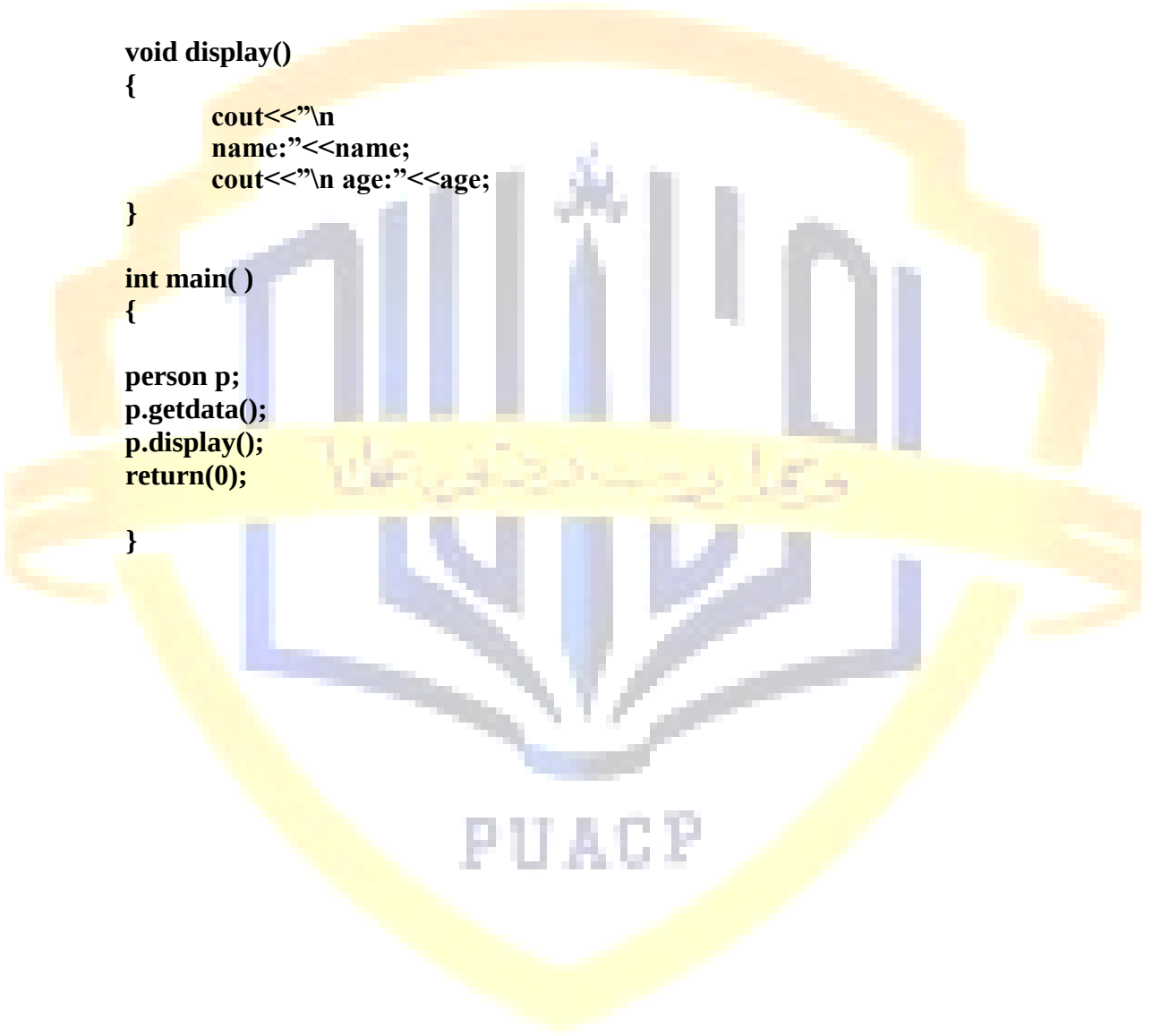
        void display(void);
    };

    void person :: getdata ( void )
    {
        cout<<"enter name";
        cin>>name;
        cout<<"enter age";
        cin>>age;
    }

    void display()
    {
        cout<<"\n
        name:"<<name;
        cout<<"\n age:"<<age;
    }

    int main()
    {
        person p;
        p.getdata();
        p.display();
        return(0);
    }

```



The smallest individual units in program are known as **tokens**. C++ has the following tokens.

- i. Keywords
- ii. Identifiers
- iii. Constants
- iv. Strings
- v. Operators

KEYWORDS:

The keywords implement specific C++ language feature. They are explicitly reserved identifiers and can't be used as names for the program variables or other user defined program elements. The keywords not found in ANSI C are shown in red letter.

C++ KEYWORDS:

asm double new switch

auto else operator template

break enum private this

case extern protected throw

catch float public try

char for register typedef

class friend return union

const goto short unsigned

continue if signed virtual

default inline sizeof void

delete long struct while

IDENTIFIERS:

Identifiers refers to the name of variable , functions, array, class etc. created by programmer. Each language has its own rule for naming the identifiers.

The following rules are common for both C and C++.

1. Only alphabetic chars, digits and under score are permitted.
2. The name can't start with a digit.
3. Upper case and lower case letters are distinct.

4. A declared keyword can't be used as a variable name.

In ANSI C the maximum length of a variable is 32 chars but in c++ there is no bar.



BASIC DATA TYPES IN C++

C ++ Data Types

User defined type Built in types Derived type
~~Structure Array~~
Union Function
Class pointer
enumeration

Integral type void Floating point

int char float double

Both C and C++ compilers support all the built in types. With the exception of void the basic datatypes may have several modifiers preceding them to serve the needs of various situations. The modifiers signed, unsigned, long and short may applied to character and integer basic data types. However the modifier long may also be applied to double.

Data types in C++ can be classified under various categories.

TYPE BYTES RANGE

char 1 -128 to – 127

unsigned 1 0 to 265

sgned char 1 -128 to 127

int 2 -32768 to 32768

unsigned int 2 0 to 65535

singed int 2 -32768 to 32768

short int 2 -32768 to 32768

long int 4 -2147483648 to 2147483648

signed long int 4 -2147483648 to 2147483648

unsigned long int 4 0 to 4294967295

float 4 3.4E-38 to 3.4E+38

double 8 1.7E -308 to 1.7E +308

long double 10 3.4E-4932 to 1.1E+ 4932

The type void normally used for:

- 1) To specify the return type of function when it is not returning any value.
- 2) To indicate an empty argument list to a function.

Example:

Void function(void);

Another interesting use of void is in the declaration of generic pointer

Example:

Void *gp;

Assigning any pointer type to a void pointer without using a cast is allowed in both C and ANSI C. In ANSI C we can also assign a void pointer to a non-void pointer without using a cast to non void pointer type. This is not allowed in C ++.

Example:

void *ptr1;

void *ptr2;

Are valid statement in ANSI C but not in C++. We need to use a cast operator.

ptr2=(char *) ptr1;

USER DEFINED DATA TYPES:

STRUCTURES AND CLASSES

We have used user defined data types such as struct, and union in C. While these more features have been added to make them suitable for object oriented programming, C++ also permits us to define

another user defined data type known as class which can be used just like any other basic data type to declare a variable. The class variables are known as objects, which are the central focus of oops.

ENUMERATED DATA TYPE:

An enumerated data type is another user defined type which provides a way for attaching names to number, these by increasing comprehensibility of the code. The enum keyword automatically enumerates a list of words by assigning them values 0,1,2 and soon. This facility provides an alternative means for creating symbolic.

Example:

```
enum shape { circle,square,triangle}
```

```
enum colour{red,blue,green,yellow}
```

```
enum position {off,on}
```

The enumerated data types differ slightly in C++ when compared with ANSI C. In C++, the tag names shape, colour, and position become new type names. That means we can declare new variables using the tag names.

Example:

```
Shape ellipse;//ellipse is of type shape
```

```
colour background ; // back ground is of type colour
```

ANSI C defines the types of enums to be ints. In C++,each enumerated data type retains its own separate type. This means that C++ does not allow an int value to be automatically converted to an enum.

Example:

```
colour background =blue; //valid
```

```
colour background =7; //error in c++
```

```
colour background =(colour) 7;//ok
```

How ever an enumerated value can be used in place of an int value.

Example:

```
int c=red ;//valid, colour type promoted to int
```

By default, the enumerators are assigned integer values starting with 0 for the first enumerator, 1 for the second and so on. We can also write

```
enum color {red, blue=4,green=8};
```

```
enum color {red=5,blue,green};
```

C++ also permits the creation of anonymous enums (i.e, enums without tag

names) Example:

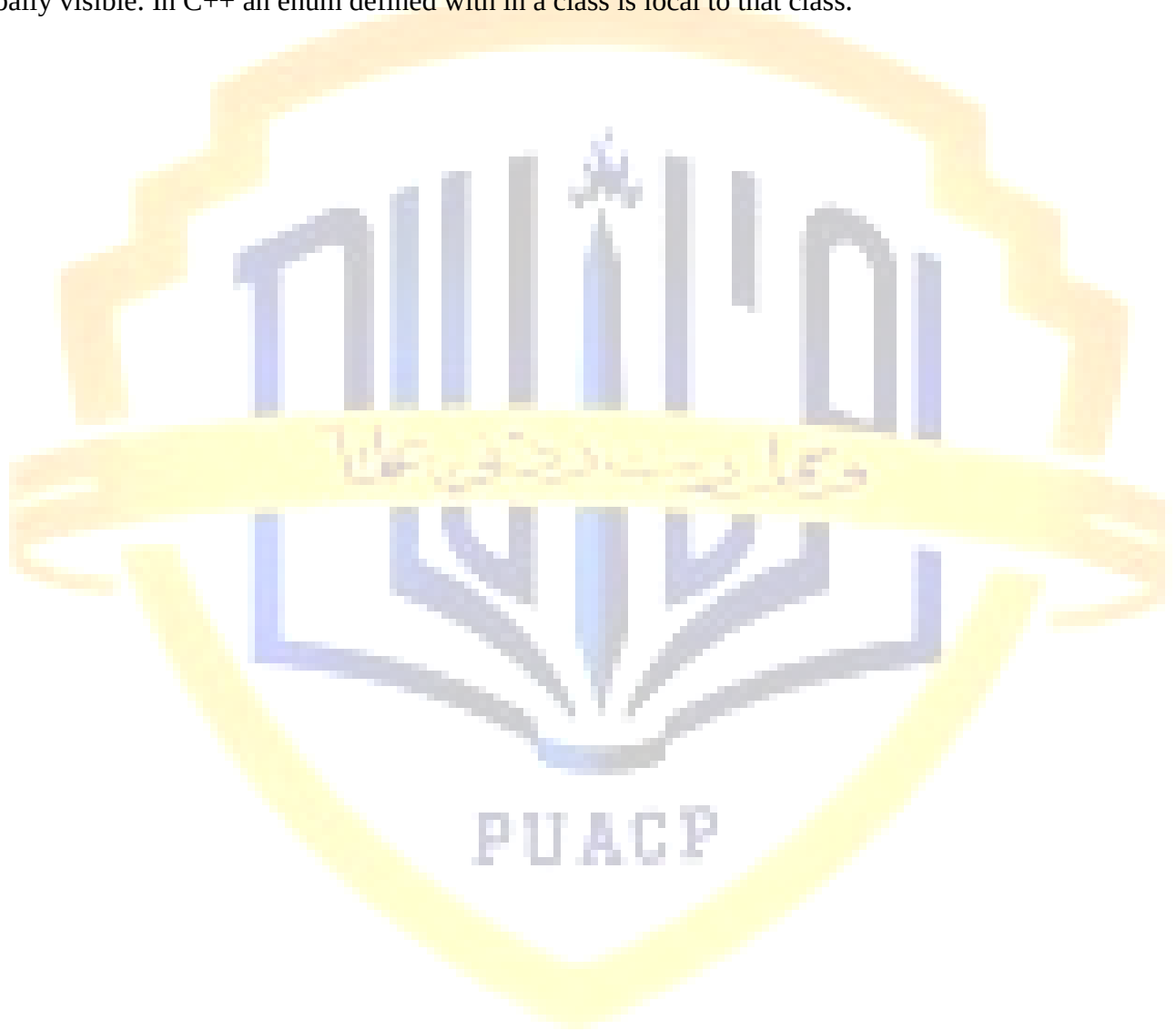
enum{off,on};

Here off is 0 and on is 1. these constants may be referenced in the same manner as regular constants. Example:

int switch-1=off;

int switch-2=on;

ANSI C permits an enum defined with in a structure or a class, but the enum is globally visible. In C++ an enum defined with in a class is local to that class.



SYMBOLIC CONSTANT:

There are two ways of creating symbolic constants in c++.

1. using the qualifier const.
2. defining a set of integer constants using enum keywords.

In both C and C++, any value declared as const can't be modified by the program in any way. In C++, we can use const in a constant expression. Such as

```
const int size = 10 ;  
char name (size) ;
```

This would be illegal in C. const allows us to create typed constants instead of having to use #define to create constants that have no type information.

```
const size=10;
```

Means

```
const int size =10;
```

C++ requires a const to be initialized. ANSI C does not require an initializer, if none is given, it initializes the const to 0.

In C++ const values are local and in ANSI C const values are global .However they can be made local made local by declaring them as static .In C++ if we want to make const value as global then declare as extern storage class.

Ex: external const total=100; Another method of naming integer constants is as follows:-

```
enum {x,y,z};
```

DECLARATION OF VARIABLES:

In ANSIC C all the variable which is to be used in programs must be declared at the beginning of the program .But in C++ we can declare the variables any where in the program where it requires .This makes the program much easier to write and reduces the errors that may be caused by having to scan back and forth. It also makes the program easier to understand because the variables are declared in the context of their use.

Example:

```
main()  
{  
    float x,average;  
    float sum=0;  
  
    for(int i=1;i<5;i++)  
    {
```

```

        cin>>x;
        sum=sum+x
    }
    float average;
        average=sum/x;
    cout<<average;
}

```

REFERENCE VARIABLES:

C++ interfaces a new kind of variable known as the reference variable. A reference variable provides an alias.(alternative name) for a previously defined variable. For example ,if we make the variable sum a reference to the variable total, then sum and total can be used interchangeably to represent the variable.

A reference variable is created as follows:

Syntax: Datatype & reference –name=variable name;

Example:

```

float total=1500;
float &sum=total;

```

Here sum is the alternative name for variable total, both the variables refer to the same data object in the memory .

A reference variable must be initialized at the time of declaration .

Note that C++ assigns additional meaning to the symbol & here & is not an address operator .The notation float & means reference to float.

Example:

```

int n[10];
int &x=n[10];
char &a='\n';

```

LECTURE-9

OPERATORS IN C++ :

C++ has a rich set of operators. All C operators are valid in C++ also. In addition, C++ introduces some new operators.

<< insertion operator

>> extraction operator

:: scope resolution operator

::* pointer to member declarator

* pointer to member operator

.* pointer to member operator

Delete memory release operator

Endl line feed operator

New memory allocation operator

Setw field width operator

SCOPE RESOLUTION OPERATOR:

Like C, C++ is also a block-structured language. Block-structured language. Blocks and scopes can be used in constructing programs. We know some variables can be declared in different blocks because the variables declared in blocks are local to that function.

Blocks in C++ are often nested.

Example:

```
-----  
-----  
{  
    Int x =10;  
    -----  
    -----  
    {  
        -----  
        -----  
        Int x=1; Block2  
        ----- Block1  
        -----  
    }  
    -----  
    -----  
}
```

Block2 contained in block 1. Note that declaration in an inner block hides a declaration of the same variable in an outer block and therefore each declaration of x causes it to refer to a different data object. Within the inner block the variable x will refer to the data object declared there in.

In C, the global version of a variable can't be accessed from within the inner block. C++ resolves this problem by introducing a new operator :: called the scope resolution operator. This can be used to uncover a hidden variable.

Syntax: :: variable –name;

Example:

```
#include <iostream.h>
int m=10;
main()
{
    int m=20;
    {
        int k=m;
        int m=30;
        cout<<"we are in inner block";
        cout<<"k="<<k<<endl;
        cout<<"m="<<m<<endl;
        cout<<":: m="<<:: m<<endl;
    }
    cout<<"\n we are in outer block \n";
    cout<<"m="<<m<<endl;
    cout<<":: m="<<:: m<<endl;
}
```

Memory Management Operator

C uses malloc and calloc functions to allocate memory dynamically at run time. Similarly it uses the functions free() to free dynamically allocated memory. We use dynamic allocation techniques when it is not known in advance how much of memory space is needed.

C++ also supports those functions; it also defines two unary operators new and delete that perform the task of allocating and freeing the memory in a better and easier way.

The new operator can be used to create objects of any type. **Syntax: pointer**

variable =new datatype;

Example:

```
p=new int; q=new int;
```

Where p is a pointer of type int and q is a pointer of type float.

```
int *p=new int;
```

```
float *p=new float;
```

Subsequently, the statements

```
*p=25;
```

`*q=7.5;`

Assign 25 to the newly created int object and 7.5 to the float object. We can also initialize the memory using the new operator.

Syntax:

```
int *p=new int(25);
float *q=new float(7.5);
```

new can be used to create a memory space for any data type including user defined such as arrays, structures, and classes. The general form for a one-dimensional array is:

```
pointer-variable=new data types [size];
```

creates a memory space for an array of 10 integers.

If a data object is no longer needed, it is destroyed to release the memory space

for reuse. **Syntax: delete pointer-variable;**

Example:

```
delete p;
delete q;
```

If we want to free a dynamically allocated array, we must use the following form of delete.

```
delete [size] pointer-variable;
```

or

```
delete [ ] pointer variable;
```

MANIPULATORS:

Manipulators are operators that are used to format the data display. The most commonly manipulators are endl and setw.

The endl manipulator, when used in an output statement, causes a line feed to be inserted. (just like \n)

Example:

```
cout<<"m="<<m<<endl;
cout<<"n="<<n<<endl;
cout<<"p="<<p<<endl;
```

If we assume the values of the variables as 2597, 14 and 175 respectively

```
m=2597; n=14;
```

```
p=175
```

If we want to print all nos in right justified way use setw which specifies a common field width for all the nos.

Example: `cout<<setw(5)<<sum<<endl;`

```
cout<<setw(10)<<"basic"<<setw(10)<<basic<<endl;
```

```
Cout<<setw(10)<<"allowance"<<setw(10)<<allowance<<endl;
```

```
cout<<setw(10)<<"total="<<setw(10)<<total;
```

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LECTURE-10

CONTROL STRUCTURES:

Like c,c++, supports all the basic control structures and implements them various control statements. **The if statement:**

The if statement is implemented in two forms:

1. simple if statement
2. if... else statement

Simple if statement:

if (condition)

{

Action;

}

If.. else statement

If (condition)

Statment1

Else

Statement2

The switch statement

This is a multiple-branching statement where, based on a condition, the control is transferred to one of the many possible points;

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Switch(expr)

{

case 1:

action1;

break;

case 2:

action2;

break;

..

..

default:

message

}

The while

statement: Syn:

While(conditi

on) {

Stements

}

The do-while statement:

Syn:

do

{

Statements

} while(condition);

The for loop:

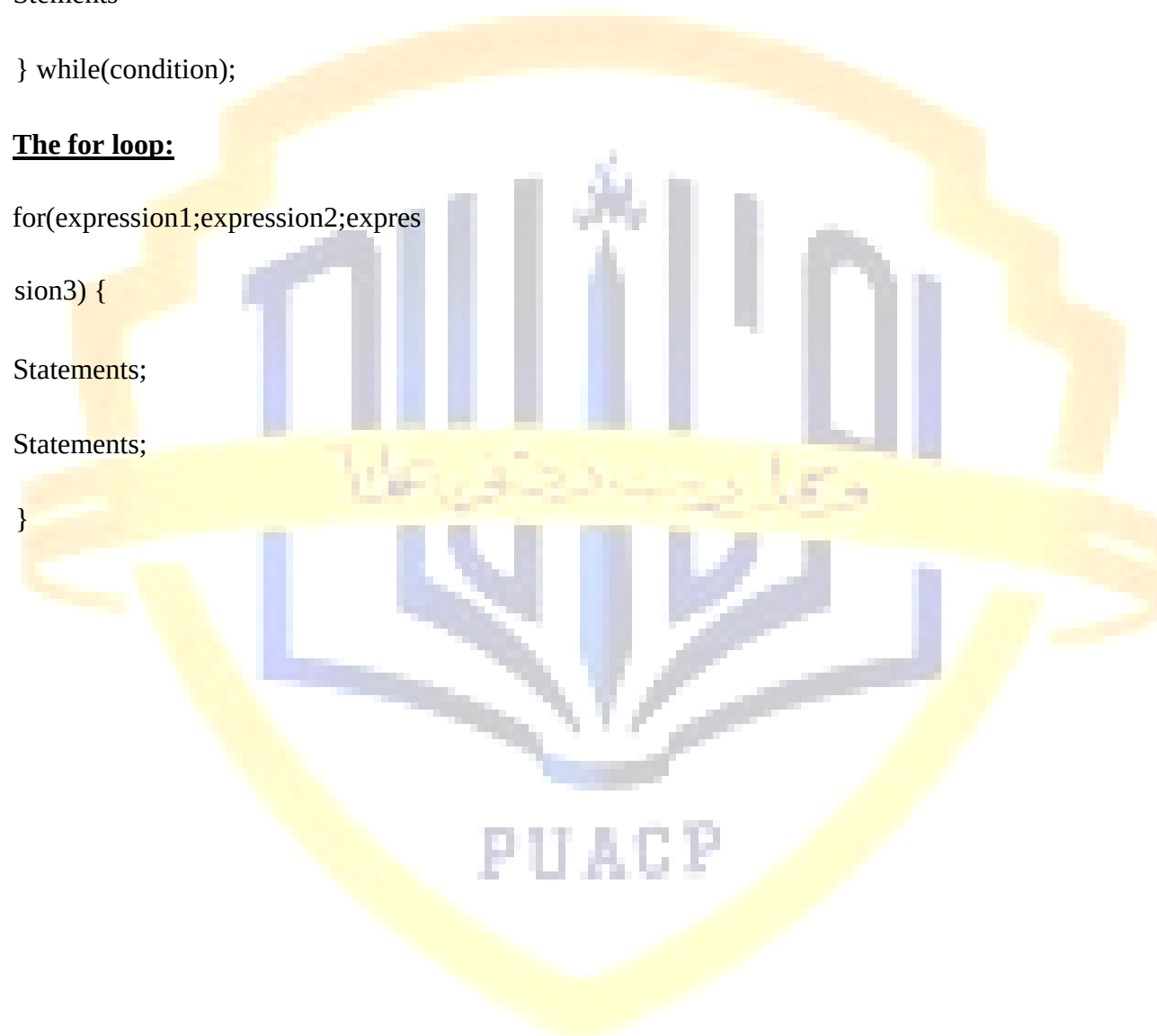
for(expression1;expression2;expres

sion3) {

Statements;

Statements;

}



LECTURE-11

FUNCTION IN C++:

The main() Function :

ANSI does not specify any return type for the main () function which is the starting point for the execution of a program .The definition of main() is :-

```
main()
{
//main program statements
}
```

This is property valid because the main () in ANSI C does not return any value. In C++, the main () returns a value of type int to the operating system. The functions that have a return value should use the return statement for terminating. The main () function in C++ is therefore defined as follows.

```
int main( )
{
-----
return(0)
}
```

Since the return type of functions is int by default, the key word int in the main() header is optional. **INLINE FUNCTION:**

To eliminate the cost of calls to small functions C++ proposes a new feature called inline function. An inline function is a function that is expanded inline when it is invoked .That is the compiler replaces the function call with the corresponding function code.

The inline functions are defined as follows:-

```
inline function-header
{
function body;
}
```

Example: inline double cube (double a)

```
{
return(a*a*a);
}
```

The above inline function can be invoked by statements like

```
c=cube(3.0);
d=cube(2.5+1.5);
```

remember that the inline keyword merely sends a request, not a command to the compiler. The compiler may ignore this request if the function definition is too long or too complicated and compile the function as a normal function.

Some of the situations where inline expansion may not work are:

1. For functions returning values if a loop, a switch or a go to exists.
2. for function s not returning values, if a return statement exists.
3. if functions contain static variables.
4. if inline functions are recursive,.

Example:

```
#include<iostream.h>
#include<stdio.h>
inline float mul(float x, float y)
{
    return(x*y);
}
inline double div(double p,double q)
{
    return(p/q);
}

main( )
{
    float a=12.345;
    float b=9.82;
    cout<<mul(a,b)<<endl;
    cout<<div (a,b)<<endl;
}
```

output:-

```
121.227898
1.257128
```

DEFAULT ARGUMENT:-

C++ allows us to call a function with out specifying all its arguments. In such cases, the function assigns a default value to the parameter which does not have a matching arguments in the function call. Default values are specified when the function is declared. The compiler looks at the prototype to see how many arguments a function uses and alerts the program for possible default values.

Example: float amount (float principle, int period ,float rate=0.15);

The default value is specified in a manner syntactically similar to a variable initialization. The above prototype declares a default value of 0.15 to the argument rate. A subsequent function call like

```
value=amount(5000,7); //one argument missing
```

passes the value of 5000 to principle and 7 to period and then lets the function, use default value of 0.15 for rate.

```
The call:- value=amount(5000,5,0.12);
```

```
//no missing argument passes an explicit value of 0.12 rate.
```

One important point to note is that only the trailing arguments can have default values. That is, we must add default from right to left. We cannot provide a default to a particular argument in the middle of an argument list.

```
Example:- int mul(int i, int j=5,int k=10); //illegal
int mul(int i=0,int j,int k=10); //illegal
int mul(int i=5,int j); //illegal
int mul(int i=2,int j=5,int k=10); //illegal
```

Default arguments are useful in situation whose some arguments always have the some value. For example, bank interest may retain the same for all customers for a particular period of

deposit.

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Example:

```
#include<iostream.h>
#include<stdio.h>
mainQ
{
float amount;
float value(float p,int n,float r=0.15);
void printline(char ch='*',int len=40);
printline( );
amount=value(5000.00,5);
cout<<"\n final value="<<amount<<endl;
printline('=');
//function definitions
float value (float p,int n, float r)
{
float si;
si=(p*n*r)/100;

return(si);
}
void printline (char ch,int len)
{
for(inti=1;i<=len;i++)
cout<<ch<<endl;
}
```

output:-

final value=10056.71613

=====

Advantage of providing the default arguments are:

1. We can use default arguments to add new parameters to the existing functions.
2. Default arguments can be used to combine similar functions into one.

CONST ARGUMENT:-

In C++, an argument to a function can be declared as const as shown below.

```
int strlen(const char *p);
int length(const string &s);
```

The qualifier const tells the compiler that the function should not modify the argument .the compiler will generate an error when this condition is violated .This type of declaration is significant only when we pass arguments by reference or pointers.

LECTURE-12

FUNCTION OVERLOADING:

Overloading refers to the use of the same thing for different purposes . C++ also permits overloading functions .This means that we can use the same function name to creates functions that perform a variety of different tasks. This is known as function polymorphism in oops.

Using the concepts of function overloading , a family of functions with one function name but with different argument lists in the functions call .The correct function to be invoked is determined by checking the number and type of the arguments but not on the function type.

For example an overloaded add() function handles different types of data as shown below.

//Declaration

```
int add(int a, int b); //prototype 1
int add (int a, int b, int c); //prototype 2
double add(double x, double y); //prototype 3
double add(double p , double q); //prototype 4
```

//function call

```
cout<<add(5,10); //uses prototype 1
cout<<add(15,10.0); //uses prototype 4
cout<<add(12.5,7.5); //uses prototype 3
cout<<add(5,10,15); //uses prototype 2
cout<<add(0.75,5); //uses prototype 5
```

A function call first matches the prototype having the same no and type of arguments and then calls the appropriate function for execution.

The function selection invokes the following steps:-

- a) The compiler first tries to find an exact match in which the types of actual arguments are the same and use that function .
- b) If an exact match is not found the compiler uses the integral promotions to the actual arguments such as :

```
char to int
float to double
to find a match
```

c)When either of them fails ,the compiler tries to use the built in conversions to the actual arguments and then uses the function whose match is unique . If the conversion is possible to have multiple matches, then the compiler will give error message.

Example:

```
long square (long n);
```

```
double square(double x);
```

A function call such as :- square(10)

Will cause an error because int argument can be converted to either long or double .There by creating an ambiguous situation as to which version of square()should be used.

PROGRAM

```
#include<iostream.h>
int volume(double,int);
double volume( double , int );
double volume(longint ,int ,int);
main( )
{
    cout<<volume(10)<<endl;
    cout<<volume(10)<<endl; cout<<volume(10)<<endl;
}
int volume( ini s)
{
    return (s*s*s); //cube
}
double volume( double r, int h)
{
    return(3.1416*r*r*h); //cylinder
}
long volume (longint l, int b, int h)
{
    return(1*b*h); //cylinder
}
```

output:- 1000
157.2595
112500

Module-2:

LECTURE-13

CLASS:-

Class is a group of objects that share common properties and relationships. In C++, a class is a new data type that contains member variables and member functions that operate on the variables. A class is defined with the keyword `class`. It allows the data to be hidden, if necessary, from external use. When we define a class, we are creating a new abstract data type that can be treated like any other built-in data type.

Generally a class specification has two parts:-

- a) Class declaration
- b) Class function definition

The class declaration describes the type and scope of its members. The class function definition describes how the class functions are implemented.

Syntax:-

```
class class-name
{
    private:
        variable declarations;
        function declaration;
    public:
        variable declarations;
        function declaration;
};
```

The members that have been declared as private can be accessed only from within the class. On the other hand, public members can be accessed from outside the class also. The data hiding is the key feature of OOPS. The use of the keyword `private` is optional by default; the members of a class are private.

The variables declared inside the class are known as data members and the functions are known as member functions. Only the member functions can have access to the private data members and private functions. However, the public members can be accessed from outside the class. The binding of data and functions together into a single class type variable is referred to as encapsulation.

Syntax:-

```
class item
{
    int member;
    float cost;
    public:
        void getldata (int a ,float b);
        void putdata (void);
};
```

The class `item` contains two data members and two function members; the data members are private by default while both the functions are public by declaration. The function `getldata()` can be used to assign values to the member variables `member` and `cost`, and `putdata()` for displaying their values. These functions provide the only access to the data members from outside the class.

CREATING OBJECTS:

Once a class has been declared we can create variables of that type by using the class

name.

Example:

item x;

creates a variable x of type item. In C++, the class variables are known as objects. Therefore x is called an object of type item.

item x, y, z also possible.

class item

{

}x, y, z;

would create the objects x, y, z of type item.

ACCESSING CLASS MEMBER:

The private data of a class can be accessed only through the member functions of that class. The main() cannot contain statements that access number and cost directly. Syntax:

object name.function-name(actual arguments);

Example:- x. getdata(100,75.5);

It assigns value 100 to number, and 75.5 to cost of the object x by implementing the getdata() function.

similarly the statement

x. putdata (); //would display the values of data members.

x. number = 100 is illegal. Although x is an object of the type item to which number belongs, the number can be accessed only through a member function and not by the object directly. Example:

class xyz

{

int x;

int y;

public:

int z;

};

xyz p;

p. x =0; error. x is private

p. z=10; ok, z is public

LECTURE-14

DEFINING MEMBER FUNCTION:

Member can be defined in two places

- Outside the class definition
- Inside the class function

OUTSIDE THE CLASS DEFINATION;

Member function that are declared inside a class have to be defined separately outside the class. Their definition are very much like the normal functions.

An important difference between a member function and a normal function is that a member function incorporates a membership. Identify label in the header. The 'label' tells the compiler which class the function belongs to.

Syntax:

```
return type class-name::function-name(argument declaration )
{
    function-body
}
```

The member ship label class-name :: tells the compiler that the function function - name belongs to the class class-name . That is the scope of the function is restricted to the class name specified in the header line. The :: symbol is called scope resolution operator.

Example:

```
void item :: getdata (int a , float b )
{
    number=a;
    cost=b;
}

void item :: putdata ( void)
{
    cout<<"number="<<number<<endl;
    cout<<"cost="<<cost<<endl;
}
```

The member function have some special characteristics that are often used in the program development.

- Several different classes can use the same function name. The "membership label" will resolve their scope, member functions can access the private data of the class .A non member function can't do so.
- A member function can call another member function directly, without using the dot operator.

INSIDE THE CLASS DEFINATION:

Another method of defining a member function is to replace the function declaration by the actual function definition inside the class .

Example:

```
class item
{
    Intnumber;
    float cost;
public:
    void getdata (int a ,float b);
    void putdata(void)
    {
        cout<<number<<endl; cout<<cost<<endl;
    }
}
```

```

    }
};

```

A C++ PROGRAM WITH CLASS:

```

#include< iostream. h>
class item
{
    int number;
    float cost;
public:
    void getdata ( int a , float b);
    void putdala ( void)
    {
        cout<<"number:"<<number<<endl;
        cout<<"cost : "<<cost<<endl;
    }
};

void item :: getdata (int a , float b)
{
    number=a;
    cost=b;
}

main ( )
{
    item x;
    cout<<"\nobjectx"<<endl;
    x. getdata( 100,299.95);
    x .putdata();
    item y;
    cout<<"\n object y"<<endl;
    y. getdata(200,175.5);
    y. putdata();
}

```

Output: object x

number 100

cost=299.950012

object -4

cost=175.5

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Q.

Write a simple program using class in C++ to input subject mark and prints it.

ans:

```

class marks
{
    private :

```

```

        int m1,m2;
        public:
        void getdata();
        void displaydata();
};
void marks :getdata()
{
    cout<<"enter 1st subject mark:";
    cin>>m1;
    cout<<"enter 2nd subject mark:";
    cin>>m2;
}
void marks :displaydata()
{
    cout<<"1st subject mark:"<<m1<<endl ;
    cout<<"2nd subject mark:"<<m2;
}
void main()
{
    clrscr();
    marks x;
    x.getdata();
    x.displaydata();
}

```

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LECTURE-15

NESTING OF MEMBER FUNCTION;

A member function can be called by using its name inside another member function of the same class. This is known as nesting of member functions.

```

#include <iostream.h>
class set
{

```

```

int m,n;
public:
    void input(void);
    void display (void);
    void largest(void);
};
int set::largest (void)
{
    if(m>n)
        return m;
    else
        return n;
}
void set::input(void)
{
    cout<<"input values of m and n:";
    cin>>m>>n;
}
void set::display(void)
{
    cout<<"largestvalue="<<largest()<<"n";
}
void main()
{
    set A;
    A.input( );
    A.display( );
}

```

output:

Input values of m and n:

3017

largest value= 30

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Private member functions:

Although it is a normal practice to place all the data items in a private section and all the functions in public, some situations may require contain functions to be hidden from the outside calls. Tasks such as deleting an account in a customer file or providing increment to an employee are events of serious consequences and therefore the functions handling such tasks should have restricted access. We can place these functions in the private section.

A private member function can only be called by another function that is a member of its class. Even an object can not invoke a private function using the dot operator.

Class sample

```
{  
    int m;  
    void read (void);  
    void write (void);  
};
```

if si is an object of sample, then

s.read();

is illegal. However the function read() can be called by the function update () to update the value of m.

```
void sample :: update(void)  
{  
    read( );  
}
```

```
#include<iostream.h>  
  
class part  
{  
private:  
    int modelnum,partnum;  
    float cost;  
public:  
    void setpart ( int mn, int pn ,float c)  
    {  
        modelnum=mn;  
        partnum=pn;  
        cost=c;  
    }  
    void showpart ( )
```

```

{
Cout<<endl<<"model:"<<modelnum<<endl;
Cout<<"num:"<< partnum <<endl
Cout<<"cost:"<<"$"<<cost;
}
};
void main()
{
part pl,p2;
p1.setpart(644,73,217.55);
p2.setpart(567,89,789.55);
pl.showpart();
pl.showpart();
}

```

output:- model:644
num:73
cost: \$217550003
model: 567
num:89
cost: \$759.549988

```

#include<iostream.h>
class distance
{
private:
    int feet;
    float inches;
public:
    void setdist ( int ft, float in)
    {
        feet=ft;
        inches=in;
    }
    void getdist()

```

```

{
cout<<"enter feet:";
cin>>feet;
cout<<"enter inches:";
cin>>inches;
}
void showdist()
{
cout<< feet<<"_"<<"inches<<endl;
}
};
void main( )
{
distance d1,d2;
d1.setdist(11,6.25);
d2.getdata();
cout<<endl<<"dist:"<<d1.showdist();
cout<<"\n"<<"dist2:";
d2.showdist();
}

```

output:- enter feet: 12

enter inches: 6.25

dist 1:"11' - 6.1.5" dist 2: 12' - 6.25"

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LECTURE-16

ARRAY WITH CLASSES:

```

#include<iostream.h>
#include<conio.h>
class employee
{
private:

```

```

        char name[20];
        int age,sal;
    public:
        void getdata();
        void putdata();
    };

void employee :: getdata ()
{
    cout<<"enter name :";
    cin>>name;
    cout<<"enter age :";
    cin>>age;
    cout<<"enter salary:";
    cin>>sal;
    return(0);
}

void employee :: putdata ( )
{
    cout<<name <<endl;
    cout<<age<<endl;
    cout<<sal<<endl;
    return(0);
}

int main()
{

```

```

    employee emp[5];
    for( int i=0;i<5;i++)
    {
        emp[i].getdata();
    }
    cout<<endl;
    for(i=0;i<5;i++)
    {
        emp[i].putdata();
    }

```

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```
    getch();  
    return(0);  
}
```

ARRAY OF OBJECTS:-

```
#include<iostream.h>  
#include<conio.h>  
class emp  
{  
    private:  
        char name[20];  
        int age,sal;  
    public:  
        void getdata( );  
        void putdata( );  
};  
void emp :: getdata( )  
{  
    cout<<"enter empname": .  
    cin>>name;  
    cout<<"enter age:"<<endl;  
    cin>>age;  
    cout<<"enter salun :";  
    cin>>sal;  
}  
void emp :: putdata ( )  
{  
    cout<<"emp name:"<<name<<endl;  
    cout<<"emp age:"<<age<<endl;  
    cout<<"emp salary:"<<sal;  
}  
  
void main()  
{
```

```

emp foreman[5];
emp engineer[5];
for(int i=0;i<5;i++)
{
cout<<" for foreman:";
foreman[i] . getdata();
}
cout<<endl;
for(i=0;i<5;i++)
{
Foreman[i].putdata(); .
}
for(int i=0;i<5;i++)
{
cout<<" for engineer:";
ingineer[i].getdata();
}
for(i=0;i<5;i++)
{
ingineer[i].putdata();
}
getch();
return(0);
}

```

REPLACE AND SORT USING CLASS:-

```

#include<iostream.h>
#include<constream.h>

class sort
{
private:
    int nm[30];
public:
    void getdata();
    void putdata();
};

void sort :: getdata()

```

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```

{
    int i,j,k;
    cout<<"enter 10 nos:" ;
    for(i=0;i<10;i++)
    {
        cin>>nm[i];
    }
    for(i=0;i<9;i++)
    {
        for(j=i+1;j<10;j++)
        {
            if(nm[i]>nm[j])
            {
k=nm[i]; nm[i]=nm[j];
nm[j]=k; }

            }
        }
    }

void sort :: putdata()
{
    int k;
    for(k=0;k<10;k++)
    {
        cout<<num [k] <<endl ;

    }
}

int main()
{
    clrscr();
    sort s;
    s.getdata();
    s.putdata();
    return(0);
}

```

ARRAY OF MEMBERS:

```

#include<iostream.h>
#include<constream.h>

const int m=50;

class items
{
    int item_code[m];

```

```

float item_price[m];
int count;

public:
void cnt(void) { count=0;}
void get_item(void);
void display_sum(void);
void remove(void);
void display _item(void);
};

void items :: get_item (void)
{
    cout<<"enter itemcode:";
    cin>> item_code[code];
    cout<<"enter item cost:";
    cin>>item_price[count];
    count ++ ;
}

void items :: display _sum(void)
{
    float sum=0;
    for( int i=0;i<count;i++)
    {
        sum=sum+item_price[i];
    }
    cout<< "\n total value:"<<sum<<endl;
}

int main ( )
{
    items order;
    order.cnt();
    int x;
    do
    {
        cout<<"\nyou can do the following:";
        cout<<"enter appropriate no:";
        cout<<endl<<" 1 :add an item";
        cout<<endl<<"2: display total value :";
        cout<<endl<<"3 : display an item";
        cout<<endl<<"4 :display all item:";
        cout<<endl<<"5 : quit:";
        cout<<endl<<endl<<"what is your option:";
    }
}

```



```

cin>>x;

switch(x)
{
    case 1: order.get_item(); break;
    case 2: order.display_sum(); break;
    case 3: order.remove(); break;
    case 4: order.display_item();break;
    case 5: break;
    default : cout<<"error in input; try again";
}
} while(x!=5);
}

```



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LECTURE-17

STATIC DATA MEMBER:

A data member of a class can be qualified as static . The properties of a static member variable are similar to that of a static variable. A static member variable has contain special characteristics.

Variable has contain special characteristics:-

- 1) It is initialized to zero when the first object of its class is created.No other initialization is permitted.
- 2) Only one copy of that member is created for the entire class and is shared by all the objects of that class, no matter how many objects are created.
- 3) It is visible only with in the class but its life time is the entire program. Static variables are normally used to maintain values common to the entire class. For example a static data member can be used as a counter that records the occurrence of all the objects.

```
int item :: count; // definition of static data member
```

Note that the type and scope of each static member variable must be defined outside the class definition. This is necessary because the static data members are stored separately rather than as a part of an object.

Example :-

```
#include<iostream.h>
class item
{
    static int count; //count is static
    int number;
public:
    void getdata(int a)
    {
        number=a;
        count++;
    }
    void getcount(void)
    {
        cout<<"count:";
        cout<<count<<endl;
    }
};

int item :: count ; //count defined

int main( )
{
    item a,b,c;
    a.get_count( );
    b.get_count( );
    c.get_count( );
    a.getdata( );
    b.getdata( );

    c.getdata( );
    cout<<"after reading data : "<<endl;
    a.get_count( );
    b.get_count( );
    c.get_count( );

    return(0);
}
```

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The output would be

```
count:0
count:0
count:0
```

After reading data

```
count: 3
count:3
count:3
```

The static Variable count is initialized to Zero when the objects created . The count is incremented whenever the data is read into an object. Since the data is read into objects three times the variable count is incremented three times. Because there is only one copy of count shared by all the three object, all the three output statements cause the value 3 to be displayed.

STATIC MEMBER FUNCTIONS:-

A member function that is declared static has following properties :-

1. A static function can have access to only other static members declared in the same class.

2. A static member function can be called using the class name as follows:-

class - name :: function - name;

Example:-

```
#include<iostream.h>
class test
{
    int code;
    static int count; // static member variable
public:
    void set(void)
    {
        code=++count;
    }
    void showcode(void)
    {
        cout<<"object member : "<<code<<end;
    }
    static void showcount(void)
    { cout<<"count="<<count<<endl; }
};

int test:: count;
int main()
{
    test t1,t2;
    t1.setcode( );
    t2.setcode( );
    test :: showcount ( ); '
    test t3;
    t3.setcode( );
    test:: showcount( );
    t1.showcode( );
    t2.showcode( );
    t3.showcode( );
    return(0);
}
```

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output:- count : 2
count: 3
object number 1
object number 2

OBJECTS AS FUNCTION ARGUMENTS

Like any other data type, an object may be used as A function argument. This can come in two ways

1. A copy of the entire object is passed to the function.
2. Only the address of the object is transferred to the function

The first method is called pass-by-value. Since a copy of the object is passed to the function, any change made to the object inside the function do not effect the object used to call the function. The second method is called pass-by-reference . When an address of the object is passed, the called function works directly on the actual object used in the call. This means that any changes made to the object inside the functions will reflect in the actual object .The pass by reference method is more efficient since it requires to pass only the address of the object and not the entire object.

Example:-

```
#include<iostream.h>
class time
{
    int hours;
    int minutes;
public:
    void gettime(int h, int m)
    {
        hours=h;
        minutes=m;
    }
    void puttime(void)
    {
        cout<< hours<<"hours and:";
        cout<<minutes<<"minutes:"<<end;
    }

    void sum( time ,time);
};

void time :: sum (time t1,time t2) . {
    minutes=t1.minutes +
    t2.minutes; hours=minutes%60;
    minutes=minutes%60;
    hours=hours+t 1.hours+t2.hours;
}

int main()
{
    time T1,T2,T3;
    T1.gettime(2,45);
    T2.gettime(3,30);
    T3.sum(T1,T2);
    cout<<"T1=";
    T1.puttime( );
    cout<<"T2=";
```

```
T2.puttime( );  
cout<<"T3=";  
T3.puttime( );  
return(0);  
}
```



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LECTURE-18

FRIENDLY FUNCTIONS:-

We know private members can not be accessed from outside the class. That is a non - member function can't have an access to the private data of a class. However there could be a case where two classes manager and scientist, have been defined we should like to use a function income tax to operate on the objects of both these classes.

In such situations, c++ allows the common function to be made friendly with both the classes, there by following the function to have access to the private data of these classes. Such a function need not be a member of any of these classes.

To make an outside function "friendly" to a class, we have to simply declare this function as a friend of the classes as shown below :

```
class ABC  
{  
-----  
-----
```

```

public:
    -----
    -----
    friend void xyz(void);
};

```

The function declaration should be preceded by the keyword friend , The function is defined else where in the program like a normal C ++ function . The function definition does not use their the keyword friend or the scope operator :: . The functions that are declared with the keyword friend are known as friend functions. A function can be declared as a friend in any no of classes. A friend function, as though not a member function , has full access rights to the private members of the class.

A friend function processes certain special characteristics:

- a. It is not in the scope of the class to which it has been declared as friend.
- b. Since it is not in the scope of the class, it cannot be called using the object of that class. It can be invoked like a member function without the help of any object.
- c. Unlike member functions.

Example:

```

#include<iostream.h>
class sample
{
    int a;
    int b;
public:
    void setvalue( ) { a=25;b=40;}
    friend float mean( sample s);
}
float mean (sample s)
{
    return (float(s.a+s.b)/2.0);
}

int main ( )
{
    sample x;
    x . setvalue( );
    cout<<"mean value="<<mean(x)<<endl;
    return(0);
}

```

output:
mean value : 32.5

A function friendly to two classes

```

#include<iostream.h>
class abc;
class xyz
{
    int x;
public:

```

```

        void setvalue(int x) { x-= I; }
        friend void max (xyz,abc);
};
class abc
{
    int a;
public:
    void setvalue( int i) {a=i; }
    friend void max(xyz,abc);
};

```

```

void max( xyz m, abc n)
{
    if(m . x >= n.a)
        cout<<m.x;
    else
        cout<< n.a;
}

```

```

int main( )
{
    abc j;
    j . setvalue( 10);
    xyz s;
    s.setvalue(20);
    max( s , j );
    return(0);
}

```

SWAPPING PRIVATE DATA OF CLASSES:

```

#include<iostream.h>

```

```

class class-2;
class class-1
{
    int value 1;
public:
    void indata( int a) { value=a; }
    void display(void) { cout<<value<<endl; }
    friend void exchange ( class-1 &, class-2 &);
};

```

```

class class-2
{
    int value2;
public:
    void indata( int a) { value2=a; }
    void display(void) { cout<<value2<<endl; }
    friend void exchange(class-1 & , class-2 &);
};
void exchange ( class-1 &x, class-2 &y)

```

```

{
    int temp=x. value 1;
    x. value I=y.valuo2;
    y.value2=temp;
}

int main( )
{
    class-1 c1;
    class-2 c2;
    c1.indata(100);
    c2.indata(200);
    cout<<"values before exchange:"<<endl;
    c1.display( );
    c2.display( );
    exchange (c1,c2);
    cout<<"values after exchange :"<< endl;
    c1. display ( );
    c2. display ( );
    return(0);
}

```

output:

```

values before exchange
100
200
values after exchange
200
100

```

PROGRAM FOR ILLUSTRATING THE USE OF FRIEND FUNCTION:

```

#include< iostream.h>
class account1;
class account2
{
private:
    int balance;
public:
    account2( ) { balance=567; }
    void showacc2( )
    {
        cout<<"balanceinaccount2 is:"<<balance<<endl;
    friend int transfer (account2 &acc2, account1 &acc1,int
        amount); };

```



```

class account1
{
private:
    int balance;
public:
    account1 ( ) { balance=345; }

    void showacc1 ( )
    {
        cout<<"balance in account1 :"<<balance<<endl;
    }
friend int transfer (account2 &acc2, account1 &acc1 ,int amount);
};

int transfer ( account2 &acc2, account1 & acc1, int amount)
{
    if(amount <=acc1 . bvalance)
    {
        acc2. balance + = amount;
        acc1 .balance - = amount;
    }
    else
        return(0);
}

int main()
{
    account1 aa;
    account2 bb;

    cout << "balance in the accounts before transfer:" ;
    aa . showacc1( );
    bb . showacc2( );
    cout << "amt transferred from account1 to account2 is:";
    cout<<transfer ( bb,aa,100)<<endl;

    cout<< " balance in the accounts after the transfer:";
    aa . showacc 1 ( );
    bb. showacc 2( );

    return(0);
}

```

output:

```

balance in the accounts before transfer
    balance in account 1 is 345
    balance in account2 is 567
and transferred from account! to account2 is 100
    balance in account 1 is 245
    balance in account2 is 667

```

LECTURE-19**RETURNING OBJECTS:**

```

#include< iostream,h>
class complex
{
    float x;
    float y;
public:
    void input( float real , float imag)
    {
        x=real;
        y=imag;
    }
    friend complex sum( complex , complex);
    void show ( complex );
};

complex sum ( complex c1, complex c2)
{
    complex c3;
    c3.x=c1.x+c2.x;
    c3.y=c1.y+c2.y;
    return c3;}

void complex :: show ( complex c)
{
    cout<<c.x<<" +j "<<c.y<<endl;
}

int main( )
{
    complex a, b,c;
    a.input(3.1,5.65);
    b.input(2.75,1.2);
    c=sum(a,b);
    cout <<" a="; a.show(a);
    cout <<" b= "; b.show(b);
    cout <<" c=" ; c.show(c);
    return(0);
}

```

output:

```

a =3.1 + j 5.65
b= 2.75+ j 1.2
c= 5.55 + j 6.85

```

POINTER TO MEMBERS:

It is possible to take the address of a member of a class and assign it to a pointer. The address of a member can be obtained by applying the operator & to a “fully qualified” class member name.

A class member pointer can be declared using the operator :: * with the class name.

For Example:

```
class A
{
private:
    int m;
public:
    void show( );
};
```

We can define a pointer to the member m as follows :

```
int A :: * ip = & A :: m
```

The ip pointer created thus acts like a class member in that it must be invoked with a class object. In the above statement. The phrase A :: * means “pointer - to - member of a class” . The phrase & A :: m means the “ Address of the m member of a class”

The following statement is not valid :

```
int *ip=&m ; // invalid
```

This is because m is not simply an int type data. It has meaning only when it is associated with the class to which it belongs. The scope operator must be applied to both the pointer and the member.

The pointer ip can now be used to access the m inside the member function (or friend function).

Let us assume that “a” is an object of “ A” declared in a member function . We can access "m" using the pointer ip as follows.

```
cout<< a . * ip;
cout<< a.m;
ap=&a;
cout<< ap-> * ip;
cout<<ap->a;
```

The deferencing operator ->* is used as to accept a member when we use pointers to both the object and the member. The dereferencing operator. .* is used when the object itself is used with the member pointer. Note that * ip is used like a member name.

We can also design pointers to member functions which ,then can be invoked using the deferencing operator in the main as shown below.

```
(object-name.* pointer-to-member function)
```

```
(pointer-to -object -> * pointer-to-member function)
```

The precedence of () is higher than that of .* and ->* , so the parenthesis are necessary.

DEREFERENCING OPERATOR:

```

#include<iostream.h>
class M
{
    int x;
    int y;
public:
    void set_xy(int a,int b)
    {
        x=a;
        y=b;
    }
friend int sum(M);
};

int sum (M m)
{
    int M :: * px= &M :: x; //pointer to member x

    int M :: * py= &m ::y; //pointer to y
    M * pm=&m;
    int s=m.* px + pm->py;
    return(s);
}

int main ( )
{
    M m;
    void(M::*pf)(int,int)=&M::set_xy; //pointer to function set-xy (n*pf)( 10,20);
    //invokes set-xy
    cout<<"sum="<<sum(n)<<endl;
    n *op=&n; //point to object n
    ( op->* pf)(30,40); // invokes set-xy
    cout<<"sum="<<sum(n)<<endl ;
    return(0);
}

```

output:

```

sum= 30
sum=70

```

CONSTRUCTOR:

A constructor is a special member function whose task is to initialize the objects of its class . It is special because its name is the same as the class name. The constructor is invoked when ever an object of its associated class is created. It is called constructor because it construct the values of data members of the class.

A constructor is declared and defined as follows:

```
//class with a constructor
class integer
{
    int m,n;
public:
    integer! void);//constructor declared
    -----
};
integer :: integer(void)
{
    m=0;
    n=0;
}
```

When a class contains a constructor like the one defined above it is guaranteed that an object created by the class will be initialized automatically.

For example:-

Integer int1; //object int 1 created

This declaration not only creates the object int1 of type integer but also initializes its data members m and n to zero.

A constructor that accept no parameter is called the default constructor. The default constructor for class A is A :: A(). If no such constructor is defined, then the compiler supplies a default constructor .

Therefore a statement such as :-

A a ;//invokes the default constructor of the compiler of the compiler to create the object "a" ;

Invokes the default constructor of the compiler to create the object a.

The constructor functions have some characteristics:-

- They should be declared in the public section .
- They are invoked automatically when the objects are created.
- They don't have return types, not even void and therefore they cannot return values.
- They cannot be inherited , though a derived class can call

the base class constructor .

- Like other C++ function , they can have default arguments,
- Constructor can't be virtual.
- An object with a constructor can't be used as a member of union.

Example of default constructor:

```
#include<iostream.h>
#include<conio.h>

class abc
{
private:
    char nm[];
public:
    abc ( )
    {
        cout<<"enter your name:";
        cin>>nm;
    }
    void display( )
    {
        cout<<nm;
    }
};

int main( )
{
    clrscr( );
    abc d;
    d.display( );
    getch( );
    return(0);
}
```

PARAMETERIZED CONSTRUCTOR:-

the constructors that can take arguments are called parameterized constructors. Using parameterized constructor we can initialize the various data elements of different objects with different values when they are created.

Example:-

```
class integer
{
    int m,n;
public:
    integer( int x, int y);
    -----
    -----
```

```
};
```

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```
integer:: integer (int x, int y)
{
    m=x;n=y;
}
```

the argument can be passed to the constructor by calling the constructor implicitly.

```
integer int 1 = integer(0,100); // explicit call
integer int 1(0,100); //implicit call
```

CLASS WITH CONSTRUCTOR:-

```
#include<iostream.h>
class integer
{
    int m,n;
public:
    integer(int,int);
    void display(void)
    {
        cout<<"m="<<m ;
        cout<<"n="<<n;
    }
};

integer :: integer( int x,int y) // constructor defined
{
    m=x;
    n=y;
}

int main( )
{
    integer int 1(0, 100); // implicit call
    integer int2=integer(25,75);
    cout<<" \nobject1 "<<<endl;
    int1.display( );
    cout<<" \n object2 "<<<endl;
    int2.display( );
}
```

output:

```
object 1
m=0
n=100
object2
m=25
n=25
```

Example:-

```
#include<iostream.h>
#include<conio.h>
class abc
{
private:
    char nm [30];
    int age;
public:
    abc ( ){ } // default
    abc ( char x[], int y);
    void get( )
    {
        cout<<"enter your name:";
        cin>>nm;
        cout<<" enter your age:";
        cin>>age;
    }
    void display( )
    {
        cout<<nm<<endl;
        cout<<age;
    }
};

abc :: abc(char x[], int y)
{
    strcpy(nm,x);
    age=y;
}

void main( )
{
    abc l;
    abc m=abc("computer",20000);
    l.get();
    l.display( );
    m.display ( );
    getch( );
}
```

OVERLOADED CONSTRUCTOR:-

```
#include<iostream.h>
#include<conio.h>
class sum
{
private:
    int a;
    int b;
    int c;
    float d;
    double e;
public:
```



```

sum ( )
{
cout<<"enter a;";
cin>>a;
cout<<"enter b;";
cin>>b;
cout<<"sum= "<<a+b<<endl;
}
sum(int a,int b);
sum(int a, float d,double c);
};
sum :: sum(int x,int y)
{
    a=x;
    b=y;
}
sum :: sum(int p, float q ,double r)
{
    a=p;
    d=q;
    e=r;
}
void main( )
{
clrscr( );
sum 1;
sum m=sum(20,50);
sum n= sum(3,3.2,4.55);
getch( );
}

output:
enter a : 3
enter b : 8
sum=11
sum=70
sum=10.75

```

COPY CONSTRUCTOR:

A copy constructor is used to declare and initialize an object from another object. Example:-

the statement

```
integer 12(11);
```

would define the object 12 and at the same time initialize it to the values of 11. Another form of this statement is : integer 12=11;

The process of initialization through a copy constructor is known as copy initialization. Example:-

```

#include<iostream.h>
class code
{
    int id;
public
    code ( ) { } //constructor

```

```

code(int a) { id=a; } //constructor
code(code &x)
{
    Id=x.id;
}
void display( )
{
    cout<<id;
}

};

int main( )
{
    code A(100);
    code B(A);
    code C=A;
    code D;
    D=A;
    cout<<"\n id of A :"; A.display( );
    cout<<"\nid of B :"; B.display( );
    cout<<"\n id of C:"; C.display( );
    cout<<"\n id of D:"; D.display( );
}

```

output :-

```

id of A:100
id of B:100
id of C:100
id of D:100

```

DYNAMIC CONSTRUCTOR:-

The constructors can also be used to allocate memory while creating objects . This will enable the system to allocate the right amount of memory for each object when the objects are not of the same size, thus resulting in the saving of memory.

Allocate of memory to objects at the time of their construction is known as dynamic constructors of objects. The memory is allocated with the help of new operator. Example:-

```

#include<iostream.h>
#include<string.h>
class string
{
    char *name;

    int length;
public:
    string ( )
    {
        length=0;
        name= new char [length+1]; /* one extra for \0 */
    }
    string( char *s) //constructor 2

```

```

    {
        length=strlen(s);
        name=new char [length+1];
        strcpy(name,s);
    }
void display(void)
{
    cout<<name<<endl;
}
void join(string &a ,string &b)
{
    length=a. length +b . length;
    delete name;
    name=new char[length+1]; /* dynamic allocation
    */ strcpy(name,a.name);
    strcat(name,b.name);
}
};
int main( )
{
    char * first = "Joseph" ;
    string name1(first),name2("louis"),name3(
    "LaGrange"),s1,s2; s1.join(name1,name2);
    s2.join(s1,name3);
    name1.display( );
    name2.display( );
    name3.display( );
    s1.display( );
    s2.display( );
}

```

output :-

```

Joseph
Louis
language
Joseph Louis
Joseph Louis Language

```

LECTURE-21

DESTRUCTOR:-

A destructor, as the name implies is used to destroy the objects that have been created by a constructor. Like a constructor, the destructor is a member function whose name is the same as the class name but is preceded by a tilde.

For Example:-

```
~ integer( ) { }
```

A destructor never takes any argument nor does it return any value. It will be invoked implicitly by the compiler upon exit from the program to clean up storage that is no longer accessible. It is a good practice to declare destructor in a program since it releases memory space for future use.

Delete is used to free memory which is created by new.

Example:-

```
matrix : : ~ matrix( )
```

```

{
    for(int i=0; i<11;i++)
        delete p[i];
    delete p;
}

```

IMPLEMENTED OF DESTRUCTORS:-

```

#include<iostream.h>
int count=0;
class alpha
{
public:
    alpha( )
    {
        count ++;
        cout<<"\n no of object created :"<<endl;
    }
    ~alpha( )
    {
        cout<<"\n no of object destroyed :"<<endl;
        count--;
    }
};

int main( )
{
    cout<<"\n \n enter main \n:";
    alpha A1,A2,A3,A4;
    {
        cout<<"\n enter block 1 :\n";
        alpha A5;
    }
    {
        cout<<"\n \n enter block2 \n";
        alpha A6;
    }
    cout<<"\n re-enter main \n:";
    return(0);
}

```

output:-

```

enter main
no of object created 1
no of object created 2
no of object created 3
no of object created 4
enter block 1
no of object created 5
no of object destroyed 5
enter block 2

```

no of object created 5
no of object destroyed 5
re-enter main
no of object destroyed 4
no of object created 3
no of object created 2
no of object created 1

Example :-

```
#include<iostream.h>
int x=1;
class abc
{
public:
    abc ( )
    {
        x--;
        cout<<"construct the no"<<x<<endl;
    }
    ~abc ( )
    {
        cout<<"destruct the no:"<<x<<endl;
        x--;
    }
};

int main ( )
{
    abc I1,I2,I3,I4;
    cout<<I1<<I2<<I3<<I4<<endl;
    return(0);
}
```

LECTURE-22 & 23

OPERATOR OVERLOADING:-

Operator overloading provides a flexible option for the creation of new definitions for most of the C++ operators. We can overload all the C++ operators except the following:

- Class members access operator (. , .*)
- Scope resolution operator (: :)
- Size operator (sizeof)
- Condition operator (? :)

Although the semantics of an operator can be extended, we can't change its syntax, the grammatical rules that govern its use such as the no of operands precedence and associativity. For example the multiplication operator will enjoy higher precedence than the addition operator.

When an operator is overloaded, its original meaning is not lost. For example, the operator +, which has been overloaded to add two vectors, can still be used to add two integers.

DEFINING OPERATOR OVERLOADING:

To define an additional task to an operator, we must specify what it means in relation to the class to which the operator is applied . This is done with the help of a special function

called operator function, which describes the task.

Syntax:-

```
return-type class-name :: operator op( arg-list)
{
    function body
}
```

Where return type is the type of value returned by the specified operation and op is the operator being overloaded. The op is preceded by the keyword operator, operator op is the function name.

operator functions must be either member function, or friend function. A basic difference between them is that a friend function will have only one argument for unary operators and two for binary operators, This is because the object used to invoke the member function is passed implicitly and therefore is available for the member functions. Arguments may be either by value or by reference.

operator functions are declared in the class using prototypes as follows:-

```
vector operator + (vector); // vector addition
vector operator-(); //unary minus
friend vector operator + (vector, vector); // vector add
friend vector operator -(vector); // unary minus
vector operator - ( vector &a); // subtraction
int operator ==(vector); //comparison
friend int operator ==(vector, vector); // comparison
```

vector is a data type of class and may represent both magnitude and direction or a series of points called elements.

The process of overloading involves the following steps:-

1. Create a class that defines the data type that is used in the overloading operation.
2. Declare the operator function operator op() in the public part of the class
3. It may be either a member function or friend function.
4. Define the operator function to implement the required operations.

Overloaded operator functions can be invoked by expressions such as

op x or x op;

for unary operators and

x op y

for binary operators.

operator op(x);

for unary operator using friend function

operator op(x,y);

for binary operator using friend function.

Unary – operator overloading(using member function):

```
class abc
```

```
{
```

```
int m,n;
```

```
public:
```

```

abc()
{
m=8;
n=9;
}
void show()
{
cout<<m<<n;
}
operator -- ()
{
--m;
--n;
}
};
void main()
{
abc x;
x.show();
--x;
x.show();
}

```

Unary – - operator overloading(using friend function):

```

class abc
{
int m,n;
public:
abc()
{
m=8;
n=9;
}
void show()
{
cout<<m<<n;
}
}

```

```

}
friend operator --(abc &p);
};

```

operator -- (abc &p)

```

{
--p.m;
--p.n;
}
};

```

```

void main()
{
abc x;
x.show();
operator--(x);
x.show();
}

```

Unary operator+ for adding two complex numbers (using member function)

```

class complex
{
float real,img;
public:
complex()
{
    real=0;
    img=0;
}
complex(float r,float i)
{
    real=r;
    img=i;
}
void show()
{
    cout<<real<<"+"<<img;
}
complex operator+(complex &p)
{
    complex w;
    w.real=real+p.real;
    w.img=img+p.img;
    return w;
}
};
void main()
{

```



```

complex s(3,4);
complex t(4,5);
complex m;
m=s+t;
s.show();
t.show();
m.show();
}

```

Unary operator+ for adding two complex numbers (using friend function)

```

class complex
{
float real,img;
public:
complex()
{
    real=0;
    img=0;
}
complex(float r,float i)
{
    real=r;
    img=i;
}
void show()
{
    cout<<real<<"+"<<img;
}
friend complex operator+(complex &p,complex &q);
};
complex operator+(complex &p,complex &q)
{
    complex w;
    w.real=p.real+q.real;
    w.img=p.img+q.img;
    return w;
}
};
void main()
{
    complex s(3,4);complex t(4,5);
    complex m;
    m=operator+(s,t);
    s.show();t.show();
    m.show();
}

```

Overloading an operator does not change its basic meaning. For example assume the + operator can be overloaded to subtract two objects. But the code becomes unreachable. class

```

integer
{
    intx, y;
    public:

```

```

        int operator + ( ) ;
    }
    int integer: : operator + ( )
    {
        return (x-y) ;
    }

```

Unary operators, overloaded by means of a member function, take no explicit argument and return no explicit values. But, those overloaded by means of a friend function take one reference argument (the object of the relevant class).

Binary operators overloaded through a member function take one explicit argument and those which are overloaded through a friend function take two explicit arguments.

Table 7.2

Operator to Overload	Arguments passed to the Member Function	Arguments passed to the Friend Function
Unary Operator	No	1
Binary Operator	1	2

LECTURE-24

Type Conversions

In a mixed expression constants and variables are of different data types. The assignment operations causes automatic type conversion between the operand as per certain rules.

The type of data to the right of an assignment operator is automatically converted to the data type of variable on the left.

Consider the following example:

```

int x;
float y = 20.123;
x=y;

```

This converts float variable y to an integer before its value assigned to x. The type conversion is automatic as far as data types involved are built in types. We can also use the assignment operator in case of objects to copy values of all data members of right hand object to the object on left hand. The objects in this case are of same data type. But of objects are of different data types we must apply conversion rules for assignment.

There are three types of situations that arise where data conversion are between incompatible types. 1. Conversion from built in type to class type.

2. Conversion from class type to built in type.

3. Conversion from one class type to another.

Basic to Class Type

A constructor was used to build a matrix object from an int type array. Similarly, we used another constructor to build a string type object from a char* type variable. In these examples constructors performed a defacto type conversion from the argument's type to the constructor's class type

Consider the following constructor:

```
string :: string (char*a)
{
    length = strlen (a);
    name=new char[len+1];
    strcpy (name,a);
}
```

This constructor builds a string type object from a char* type variable a. The variables length and name are data members of the class string. Once you define the constructor in the class string, it can be used for conversion from char* type to string type.

Example

```
string s1 , s2;
char* name1 = "Good Morning";
char* name2 = "STUDENTS";
s1 = string(name1);
s2 = name2;
```

The program statement

```
si = string (name1);
```

first converts name 1 from char* type to string type and then assigns the string type values to the object s1. The statement

```
s2 = name2;
```

performs the same job by invoking the constructor implicitly.

Consider the following example

```
class time
{
    int hours;
    int minutes;
    public:
    time (int t) // constructor
    {
        hours = t / 60; //t is inputted in minutes
        minutes = t % 60;
    }
};
```

In the following conversion statements :

```
time T1; //object T1 created
int period = 160;
T1 = period; //int to class type
```

The object T1 is created. The variable period of data type integer is converted into class type time by invoking the constructor. After this conversion, the data member hours of T1 will have value 2 and minutes will have a value of 40 denoting 2 hours and 40 minutes.

Note that the constructors used for the type conversion take a single argument whose type is to be converted.

In both the examples, the left-hand operand of = operator is always a class object. Hence, we can also accomplish this conversion using an overloaded = operator.

LECTURE-25

Class to Basic Type

The constructor functions do not support conversion from a class to basic type. C++ allows us to define a overloaded casting operator that convert a class type data to basic type. The general form of an overloaded casting operator function, also referred to as a conversion function, is: operator

```
typename ( )  
{  
    //Program statmerit .  
}
```

This function converts a class type data to typename. For example, the operator double() converts a class object to type double, in the following conversion function:

```
vector:: operator double ( )  
{  
    double sum = 0 ;  
    for(int I = 0; ioize;  
    sum = sum + v[i] * v[i ] ; //scalar magnitude  
    return sqrt(sum);  
}
```

The casting operator should satisfy the following conditions.

- It must be a class member.
- It must not specify a return type.
- It must not have any arguments. Since it is a member function, it is invoked by the object and therefore, the values used for, Conversion inside the function belongs to the object that invoked the function. As a result function does not need an argument.

In the string example discussed earlier, we can convert the object string to char* as follows: string:: operator char*()

```
{  
    return (str) ;  
}
```

One Class to Another Class Type

We have just seen data conversion techniques from a basic to class type and a class to basic type. But sometimes we would like to convert one class data type to another class type.

Example

Obj1 = Obj2 ; //Obj1 and Obj2 are objects of different classes.

Obj1 is an object of class one and Obj2 is an object of class two. The class two type data is converted to class one type data and the converted value is assigned to the Obj1. Since the conversion takes place from class two to class one, two is known as the source and one is known as the destination class.

Such conversion between objects of different classes can be carried out by either a constructor or a conversion function. Which form to use, depends upon where we want the type conversion function to be located, whether in the source class or in the destination class. We studied that the casting operator function

Operator typename()

Converts the class object of which it is a member to typename. The type name may be a built-in type or a user defined one(another class type) . In the case of conversions between objects, typename refers to the destination class. Therefore, when a class needs to be converted, a casting operator function can be used. The conversion takes place in the source class and the result is given to the destination class object.

Let us consider a single-argument constructor function which serves as an instruction for converting the argument's type to the class type of which it is a member. The argument belongs to the source class and is passed to the destination class for conversion. Therefore the conversion constructor must be placed in the destination class.

Table 7.3

Conversion	Conversion takes place in
Source class	Destination class
Basic to class	Not applicable
Class to Basic	Casting operator
Class to class	Casting operator
	Constructor

When a conversion using a constructor is performed in the destination class, we must be able to access the data members of the object sent (by the source class) as an argument. Since data members of the source class are private, we must use special access functions in the source class to facilitate its data flow to the destination class.

Consider the following example of an inventory of products in a store. One way of keeping record of the details of the products is to record their code number, total items in the stock and the cost of each item. Alternatively we could just specify the item code and the value of the item in the stock. The following program uses classes and shows how to convert data of one type to another.

```
#include<iostream.h>
#include<conio.h>
class stock2;
class stock1
{
int code, item;
float price;
public:
stock1 (int a, int b, float c)
{
code=a;
item=b;
price=c;
}
void disp( )
{
cout<<"code"<<code <<"\n";
cout<<"Items"<<item <<"\n";
cout<<"Price per item Rs . "<<price <<"\n";
}
int getcode( )
{return code; }
int getitem( )
{return item; }
int getprice( )
{return price;}
```